

Erdős Problem #404

EP-404

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

For which integers $a \geq 1$ and primes p is there a finite upper bound on those k such that there are $a = a_1 < \dots < a_n$ with $p^k \mid (a_1! + \dots + a_n!)$? If $f(a, p)$ is the greatest such k , how does this function behave?

Is there a prime p and an infinite sequence $a_1 < a_2 < \dots$ such that if p^{m_k} is the highest power of p dividing $\sum_{i \leq k} a_i!$ then $m_k \rightarrow \infty$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>